
Machine Learning A

2026-2027

Home Assignment 3

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The deadline for this assignment is **24 September 2026, 22:00**. You must submit your *individual* solution electronically via the Absalon home page.

A solution consists of:

- A PDF file with detailed answers to the questions, which may include graphs and tables if needed. Do *not* include your full source code in the PDF file. You may include short, selected code snippets if you are asked to do so or if they help improve the readability or clarity of your answers.
- Your solution source code, with comments about the major steps involved in each question (see below). If your solution consists of multiple source files, submit them together in a .zip file; if there is only one source file, it may be submitted directly without zipping. Source code must be submitted in the original file format, not as PDF. The programming language of the course is Python.
- **IMPORTANT: Do NOT zip the PDF file**, since zipped files cannot be opened in *SpeedGrader*. Zipped PDF submissions will not be graded.
- Your PDF report should be self-sufficient. That is, it should be possible to grade your submission without opening the source-code submission. In particular, all requested outputs, such as plots, tables, and numerical values, must be generated and included in the PDF report. We do not guarantee that the source-code submission will be opened during grading.
- Your code should be structured such that there is one main file (or one main file per question) that we can run to reproduce all the results presented in your report. This main file can, if you like, call other files with functions, classes, etc.

- Handwritten solutions will not be accepted. Please use the provided LaTeX template to write your report.
- You must adhere to the notations used in the task and in the lectures. If you need to use additional notations, ensure they are clearly and precisely defined. Undefined notations will result in point deduction.

1 Optimal Classification (16 points) [Christian, W3 Friday]

Assume that you have some input X taking values in $\{0, 1\}$ and a target Y also taking values in $\{0, 1\}$. The following table shows the exact values of $P(Y = y | X = x)$, which you are happy to know:

	$P(Y = y X = x)$	
	$y = 0$	$y = 1$
$x = 0$	1	0
$x = 1$	0.8	0.2

For example, $P(Y = 1 | X = 1) = 0.2$. All inputs occur with equal probability.

1. Write down the optimal deterministic classification rule mapping an input x to an output y minimizing the 0-1 loss. Analytically compute the risk, that is, the expected 0-1 loss, of this classification rule.
2. Now someone suggests the following probabilistic classifier: The probabilistic classifier assigns the label y to an input x with probability $P(Y = y | X = x)$. Analytically compute the risk, that is, the expected 0-1 loss, of this classification rule.
3. Can you find a probabilistic classifier that is better than the better of the two considered above? If yes, state a probabilistic classification rule that is better than the two classifiers considered above. If no, provide a formal proof that you cannot get better.

2 Logistic Regression (50 points) [Christian, W3 Friday]

Section 3.3 in the textbook (Abu-Mostafa et al., 2012) is a good read. You can also find a scanned version of the chapter on Absalon.

In the lecture, we encoded the labels of binary classification by $\{0, 1\}$, Abu-Mostafa et al. (2012) prefer $\{-1, 1\}$. In this assignment, we will consider both variants.

2.1 Cross-entropy error measure (16 points)

In the lecture, we derived the objective function

$$\mathcal{L}(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^N \ln \left(1 + e^{-y_n \mathbf{w}^T \mathbf{x}_n} \right) \quad (1)$$

for logistic regression using the maximum likelihood principle.

2.1.1 Rewriting negative logarithmic likelihood as cross-entropy

If we are learning from data $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)\} \subseteq (\mathbb{R}^d \times \{-1, 1\})^N$ to predict noisy targets encoded by $\{-1, 1\}$, show that the maximum (logarithmic) likelihood method in general reduces to finding the hypothesis h describing

$$h(\mathbf{x}) = P(Y = 1 \mid X = \mathbf{x})$$

which minimizes

$$\mathcal{L}(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^N \left[\mathbb{1}_{(y_n=1)} \ln \frac{1}{h(\mathbf{x}_n)} + \mathbb{1}_{(y_n=-1)} \ln \frac{1}{1 - h(\mathbf{x}_n)} \right] , \quad (2)$$

where the indicator function $\mathbb{1}_{(\cdot)}$ is 1 if its argument is true and 0 otherwise.

2.1.2 Cross-entropy and logistic regression

For the case $h(\mathbf{x}_n) = \theta(\mathbf{w}^T \mathbf{x}_n)$, argue that minimizing the error (2) is equivalent to minimizing (1).

Recall that a Bernoulli random variable with parameter p takes the value 1 with probability p and the value 0 with probability $1 - p$. For two Bernoulli random variables with parameters p and q , respectively, the *cross-entropy* from information theory is defined as

$$p \ln \frac{1}{q} + (1 - p) \ln \frac{1}{1 - q} .$$

Thus, the objective function (2) above corresponds to a cross-entropy error measure on the data point $(\mathbf{x}_n, y_n)$ with $p = \mathbb{1}_{(y_n=1)}$ and $q = h(\mathbf{x}_n)$.

2.2 Logistic regression loss gradient (24 points)

2.2.1 Gradient for -1 and 1

For data $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)\} \subseteq (\mathbb{R}^d \times \{-1, 1\})^N$, show that in the case of logistic regression we have the following gradient of the negative log-likelihood:

$$\begin{aligned}\nabla \mathcal{L}(\mathbf{w}) &= -\frac{1}{N} \sum_{n=1}^N \frac{y_n \mathbf{x}_n}{1 + e^{y_n \mathbf{w}^T \mathbf{x}_n}} \\ &= \frac{1}{N} \sum_{n=1}^N -y_n \mathbf{x}_n \theta(-y_n \mathbf{w}^T \mathbf{x}_n)\end{aligned}$$

Here θ denotes the sigmoid function.

2.2.2 Gradient for 0 and 1

Now assume labels in $\{0, 1\}$. Show that this leads to

$$\nabla \mathcal{L}(\mathbf{w}) = -\frac{1}{N} \sum_{n=1}^N [y_n - \theta(\mathbf{w}^T \mathbf{x}_n)] \mathbf{x}_n .$$

Hint: For the $\{0, 1\}$ case the lecture slides provide the answer, you just need to add an explanation and intermediate steps.

2.2.3 Influence of misclassified examples

Argue for both label encodings that a “misclassified” example contributes more to the gradient than a correctly classified example.

2.3 Log-odds (10 points)

Consider binary logistic regression. Let the input space be $\mathbb{R}^d$ and the label space be $\{0, 1\}$. Let a model f with parameters $\mathbf{w} \in \mathbb{R}^d$ and $b \in \mathbb{R}$ be defined as:

$$f(\mathbf{x}) = \sigma(\mathbf{w}^T \mathbf{x} + b) = P(Y = 1 | X = \mathbf{x}) \quad (3)$$

Prove that if the (affine) linear part of the model encodes the log-odds, that is, if

$$\mathbf{w}^T \mathbf{x} + b = \ln \frac{P(Y = 1 | X = \mathbf{x})}{P(Y = 0 | X = \mathbf{x})} , \quad (4)$$

then σ is the logistic function. That is, if $\mathbf{w}^T \mathbf{x} + b$ encodes on log-scale how frequent class 1 occurs relative to class 0, then σ is the logistic function.

3 Sleep Well (34 points) [Christian, W4 Monday]

Sleep is one of the most fundamental physiological processes, and abnormal sleeping patterns are associated with poor health. They may, for example, indicate brain- & heart diseases, obesity and diabetes. During sleep our brain goes through a series of changes between different *sleep stages*, which are characterized by specific brain and body activity patterns. *Sleep staging* refers to the process of mapping these transitions over a night of sleep. This is of fundamental importance in sleep medicine, because the sleep patterns combined with other variables provide the basis for diagnosing many sleep related disorders (Kales and Rechtschaffen, 1968, Iber and AASM, 2007).

The stages can be determined by measuring the neuronal activity in the cerebral cortex (via electroencephalography, EEG), eye movements (via electrooculography, EOG), and/or the activity of facial muscles (via electromyography, EMG) in a *polysomnography* (PSG) study. The classification into stages is done manually. This is a difficult and time-consuming process, in which expert clinicians inspect and segment the typically 8–24 hours long multi-channel signals. Contiguous, fixed-length intervals of 30 seconds are considered, and each of these *segments* is classified individually.

Algorithmic sleep staging aims at automating this process. The state-of-the-art in algorithmic sleep staging is marked by deep neural networks, which can be highly accurate and robust, even compared to human performance, see the recent work by Perslev et al. (2019) and references therein.

This assignment considers algorithmic sleep staging. The data is based on a single EEG channel from the Sleep-EDF-15 data set (Kemp et al., 2000, Goldberger et al., 2000). The input is given by an intermediate representation from the U-Time neural network architecture (Perslev et al., 2019), the targets are sleep stages.

We created a training and test set, the inputs and the corresponding labels can be found in `X_train.csv` and `y_train.csv` and `X_test.csv` and `y_test.csv`, respectively.

3.1 Data understanding and preprocessing (4 points)

Download and extract the data from <https://github.com/christian-igel/ML/tree/main/data/Sleep/>.

Consider the training data `X_train.csv` and the corresponding labels `y_train.csv`. Report the class frequencies, that is, for each of the 5 classes report the number of data points divided by the total number of data points in the training data.

The i th row in `X_train.csv` are the features of the i th training pattern. The class label of the i th pattern is given in the i th row of `y_train.csv`.

Deliverables: description of software used; frequency of classes

3.2 Classification (30 points)

The task is to evaluate several multi-class classifiers on the data. Build the models using the training data only. The test data must only be used for final evaluation.

3.2.1 Logistic regression

Apply multinomial logistic regression. If you use regularization, describe the type of regularization you used. Report training and test loss (in terms of 0-1 loss).

Deliverables: description of software used; training and test errors; description of regularization and model selection process if used

3.2.2 Random forest

Apply random forests with 50, 100, and 200 trees. Report training and test loss (in terms of 0-1 loss) as well as out-of-bag error.

Deliverables: description of software used; training and test errors; out-of-bag error; description of regularization and model selection process if used

3.2.3 Nearest neighbor

Apply k -nearest-neighbor classification. Use cross-validation to determine the number of neighbors. Report training and test loss (in terms of 0-1 loss). Describe how you determined the number of neighbors.

Deliverables: description of software used; training and test errors; description of regularization and model selection process if used

4 Overfitting (optional, 0 points) [Christian]

John Langford wrote a very true blog article: “Clever methods of overfitting,” <http://hunch.net/?p=22>, 2005.¹ Please read it and reflect on it. You get no points for this, but there are more important things than points.

¹If the link is not working, try: <http://www.kdnuggets.com/2015/01/clever-methods-overfitting-avoid.html>

References

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